

Prime squares and half-turns

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Abstract

Can a $p \times p$ square, p prime, be cut into p congruent pieces of p unit cells other than $1 \times p$ bars? The question is open when the pieces may be turned and reflected freely. We show that the answer is no when every piece is a translate of one fixed piece or of its half-turn, for every prime p and without assuming the piece is connected; for odd p the half-turn may be replaced by any one fixed reflection of the square. The proof encodes the tiling as an identity between Laurent polynomials and counts orientations by evaluating a divisibility at $(1, 1)$. The algebraic core is checked in the Lean proof assistant.

1 The question

A p -omino is a connected set of p unit cells. JetfiRex asked on Mathematics Stack Exchange and on MathOverflow [1, 2] whether a $p \times p$ square, with p prime, can be cut into p congruent p -ominoes in any way other than into p straight bars. Stop and try $p = 3$ before reading on: the only other tromino is the L, and three L's cannot fill a 3×3 square (the corner cells force it quickly).

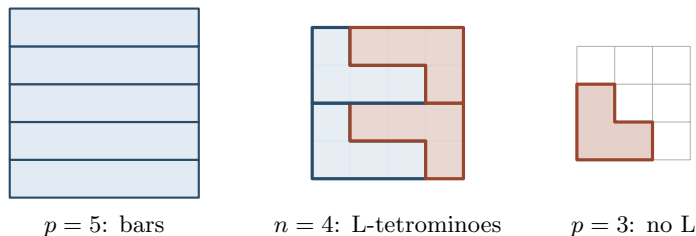


Figure 1: Left: the bar tiling, which works for every n . Middle: for $n = 4$ the L-tetromino tiles the square, and each 2×4 half is an L together with its half-turn, so composite sizes have other tilings even with only translations and half-turns. Right: the L-tromino, which cannot tile the 3×3 square.

The answer depends on primality (Figure 1). An exhaustive search, allowing all eight orientations, finds only bars for every prime $p \leq 13$; for composite $n \leq 14$ there are always other tilings. The question is a discrete case of an older one. Danzer conjectured that a square cannot be cut into five congruent polygons, convex or not, other than rectangles, and that conjecture is still open [5]. For convex pieces much more is known: the square cannot be cut into 5, 7 or 9 congruent convex pieces other than rectangles [8, 5]. Yuan, Zamfirescu and Zamfirescu asked whether the same holds for every prime number of convex pieces, and Rao, Ren and Wang conjectured it for every odd number [6]. A polyomino need not be convex, so none of this applies. For three pieces Maltby classified all ways to cut a rectangle into three congruent topological discs [4], which settles $p = 3$.

This note proves the question in a restricted form, in which the pieces may be moved by translations and half-turns but not by quarter-turns or reflections.

Theorem 1. *Let p be prime and let P be a set of p cells, not necessarily connected. If the $p \times p$ square is tiled by p copies of P , each a translate of P or of its half-turn, then P is a straight bar.*

Theorem 2. *Let p be an odd prime and let τ be one of the four reflections of the square. If the square is tiled by p copies of a p -cell set P , each a translate of P or of $\tau(P)$, then P is a straight bar.*

With translations alone the conclusion is close to known results: if translates of a set of prime size tile $\mathbb{Z}^n$, then some lattice of its translates tiles $\mathbb{Z}^n$ too, a theorem of Szegedy [7] to which Horak and Kim gave a polynomial proof [3]. We do not claim the translation-only case as new. The point here is that a second orientation can be allowed, and the proof is a page of algebra.

2 Cells as monomials

Let p be odd for now and put $m = (p-1)/2$. Index the cells of the board by their centres (i, j) with $-m \leq i, j \leq m$, and give each cell the monomial $x^i y^j$. A set of cells becomes the sum of its monomials (Figure 2). Exponents may be negative, so we work in the ring $A = \mathbb{Z}[x, x^{-1}, y, y^{-1}]$ of integer Laurent polynomials. Three features of A make it convenient.

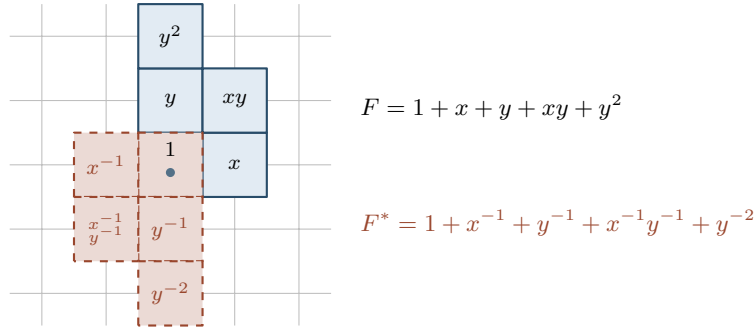


Figure 2: A P-pentomino (solid) and its half-turn about the centre (dashed), with the monomial of each cell. The half-turn is the substitution $x \mapsto x^{-1}$, $y \mapsto y^{-1}$.

Translations are units. Moving a set by (a, b) multiplies its polynomial by $x^a y^b$, and the units of A are exactly the signed monomials $\pm x^a y^b$.

The half-turn is a ring automorphism. Write $H^*(x, y) = H(x^{-1}, y^{-1})$. This is the half-turn about the centre of the board, and it respects sums and products. Each reflection of the square is likewise a substitution ($x \leftrightarrow x^{-1}$, or $y \leftrightarrow y^{-1}$, or $x \leftrightarrow y$, or $x \leftrightarrow y^{-1}$), hence an automorphism of A that squares to the identity.

Unique factorisation, and the board's two factors. The board is

$$R = x^{-m} y^{-m} Q_p(x) Q_p(y), \quad Q_p(t) = 1 + t + \cdots + t^{p-1}.$$

The ring A is a unique factorisation domain, being a localisation of $\mathbb{Z}[x, y]$. The polynomial Q_p is irreducible over $\mathbb{Z}$, since $Q_p(t+1)$ is an Eisenstein polynomial at p . A prime of $\mathbb{Z}[x]$ stays prime in $\mathbb{Z}[x][y]$, and a prime of $\mathbb{Z}[x, y]$ that does not divide xy stays prime in the localisation A ; so $Q_p(x)$ and $Q_p(y)$ are primes of A , and up to units they are the only prime factors of R . Every map in sight fixes R , and evaluation at $(1, 1)$ sends $Q_p(x)$, $Q_p(y)$ and the polynomial F of any p -cell set to p .

Now suppose the board is tiled as in Theorem 1. Let F be the polynomial of P , let U be the sum of the translation monomials used by the copies of P , and V the same for the copies of P^* .

Exact covering says

$$F U + F^* V = R. \quad (1)$$

Read it as: every cell of the board is covered exactly once, either by a translate of P or by a translate of its half-turn. If u copies have the first orientation and v the second, evaluation at $(1, 1)$ gives $U(1, 1) = u$, $V(1, 1) = v$ and $u + v = p$.

3 Proof

Everything rests on one question: do F and F^* have a common factor?

Lemma 1 (a common factor forces a bar). *If $Q_p(x)$ divides F in A , then P is a horizontal bar; likewise for $Q_p(y)$ and a vertical bar.*

Proof. Translate P so that its smallest x - and y -exponents are 0; then $F \in \mathbb{Z}[x, y]$ has degree at most $p - 1$ in x , since a copy of P fits in the board. Clearing denominators in $F = Q_p(x) G$ gives $x^a y^b F = Q_p(x) G_0$ with $G_0 \in \mathbb{Z}[x, y]$. The prime $Q_p(x)$ of $\mathbb{Z}[x, y]$ divides neither x nor y , so it divides F in $\mathbb{Z}[x, y]$, and therefore divides each row $F_k(x)$, the coefficient of y^k in F . A row has degree at most $p - 1$ and coefficients 0 or 1; a multiple of $Q_p(x)$ of that degree is $c Q_p(x)$ with c an integer, and c is the constant term, so each row is 0 or $Q_p(x)$: empty, or a full row of p cells. As P has p cells, exactly one row is full. $\square$

Proof of Theorem 1 for odd p . Suppose first that F and F^* have a common factor that is not a unit. By (1) it divides R , so one of the primes $Q_p(x)$, $Q_p(y)$ divides it, and hence divides F ; Lemma 1 gives a bar.

Otherwise F and F^* are coprime. Apply $*$ to (1); as $R^* = R$ and $**$ is the identity, $F^* U^* + F V^* = R$. Subtracting this from (1),

$$F(U - V^*) = F^*(U^* - V).$$

Since F is coprime to F^* , it divides $U^* - V$: there is $H \in A$ with $U^* - V = F H$. Evaluate at $(1, 1)$. The left side becomes $u - v$, and the right side $p H(1, 1)$, an integer multiple of p . Together with $u + v = p$ this says p divides $2u$; as p is odd, $u = 0$ or $u = p$. So all the copies have the same orientation, and (1) reads $F U = R$ or $F^* V = R$. In the second case apply $*$. Either way F divides R . As $F(1, 1) = p$ is not ± 1 , F is not a unit, so one of the primes $Q_p(x)$, $Q_p(y)$ divides F , and Lemma 1 gives a bar. $\square$

The case $p = 2$. Any two cells are symmetric about their midpoint, so the half-turn of P is a translate of P and the tiling uses translations only: $F U = R$ with $R = (1 + x)(1 + y)$, indexing cells from 0. As $F(1, 1) = 2$, F is not a unit, so $1 + x$ or $1 + y$ divides F , and Lemma 1, whose proof never used that p is odd, gives a domino.

Proof of Theorem 2. Replace the half-turn by the substitution of the chosen reflection. Nothing in the proof used more than three facts about $*$: it is a ring automorphism of A of order two, it fixes R , and it commutes with evaluation at $(1, 1)$. All three hold for each reflection. $\square$

Oddness matters in Theorem 2. For $p = 2$ the two cells on a diagonal of the 2×2 square, together with their mirror image in a vertical line, tile it. The orientation count is where the proof breaks: $p = 2$ divides $2u$ for every u .

Remark. The argument stops at quarter-turns. If ρ is a quarter-turn, applying ρ to $F U + \rho(F) V = R$ produces $\rho(F)\rho(U) + \rho^2(F)\rho(V) = R$, which involves a third tile polynomial $\rho^2(F)$, and the subtraction no longer isolates two coprime factors. Two different reflections fail in the same way. The full question needs a further idea.

Verification. Lemma 1 in polynomial form and the orientation argument are checked in Lean 4 with Mathlib [9], using only the standard axioms. The orientation argument is formalised once for any unique factorisation domain with an involutive automorphism $*$ fixing $R = q_1q_2$ (q_1, q_2 prime) and an evaluation to $\mathbb{Z}$ fixed by $*$, so it covers Theorems 1 and 2 together. Quoted rather than formalised: that A is a unique factorisation domain in which $Q_p(x)$ and $Q_p(y)$ are prime, and the passage from a tiling to (1). A Python program audits each ingredient in exact arithmetic (cell images under the eight symmetries, the substitutions, the factor obstruction, the quotients and the orientation counts, with a composite control), and deliberately wrong variants fail it. A C program searches exhaustively for $n \leq 14$ with all eight orientations, and an independent program, sharing no code with the others, enumerates the tilings with translations and one second orientation for small primes, including disconnected pieces for $p \leq 5$. Everything is at <https://github.com/jvvk/mathematics/tree/main/prime-squares-half-turns>.

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